# ClinicalTrials.gov Viz Agent - Database Schema (Postgres)

PK = primary key, FK = foreign key, UQ = unique (partial), dashed = logical reference (no FK)

**registry.term_sets (a term's trials, Q22)**

| | | |
|---|---|---|
| term_key | text | PK |
| param, expression | text | |
| size | integer | |
| data_timestamp | timestamptz | |
| resolved_at, checked_at | timestamptz | |

1 term : N trials

**registry.term_members**

| | | |
|---|---|---|
| term_key | text | PK, FK |
| nct_id | text | PK, idx |

**registry.sync_runs (loads and updates)**

| | | |
|---|---|---|
| run_id | integer | PK |
| kind | text (load\|update) | |
| status | text (running\|succeeded\|failed) | idx |
| data_timestamp | timestamptz | |
| since | date (nullable) | |
| upserted, removed | integer | |
| error | text (nullable) | |
| started_at, finished_at | timestamptz | idx |

**registry.trials (the mirror, Q22)**

| | | |
|---|---|---|
| nct_id | text | PK |
| record | jsonb (the API's study) | |
| start_year | integer (nullable) | idx |
| last_update_post_date | date (nullable) | |
| synced_at | timestamptz | |

**plan_cache (L2, standalone)**

| | | |
|---|---|---|
| request_key | char(64) | PK |
| planner_version | text | |
| plan | jsonb | |
| planner_trace | jsonb | |
| created_at | timestamptz | |
| last_used_at | timestamptz | |
| hit_count | integer | |

**api_searches (L1 cache, search level)**

| | | |
|---|---|---|
| search_key | char(64) | PK |
| endpoint | text | |
| params | jsonb | |
| kind | text (count\|pages\|stats\|study) | |
| total_count | integer | |
| pages_expected | integer | |
| complete | boolean | |
| verified_data_ts | timestamptz | FK |
| fetched_at | timestamptz | |
| last_verified_at | timestamptz | |
| last_used_at | timestamptz | |

1 search : N pages

**api_pages (raw pages)**

| | | |
|---|---|---|
| search_key | char(64) | PK, FK |
| page_index | integer | PK |
| n_studies | integer | |
| body_gz | bytea (gzip raw bytes) | |
| raw_bytes | integer | |
| fetched_at | timestamptz | |

**data_versions**

| | | |
|---|---|---|
| data_timestamp | timestamptz | PK |
| api_version | text | |
| first_seen_at | timestamptz | |
| last_checked_at | timestamptz | |

verified against

verified against

**results (L3, immutable rows)**

| | | |
|---|---|---|
| result_id | uuid | PK |
| result_key | char(64) | idx |
| status | text (ok\|no_matching_trials\|...) | |
| request | jsonb | |
| plan | jsonb | |
| response | jsonb | |
| search_keys | char(64)[] | logical |
| verified_data_ts | timestamptz | FK |
| schema_version | text | |
| code_version | text | |
| created_at | timestamptz | |
| expires_at | timestamptz | idx |

1 result : N data points

**result_datums (one per data point)**

| | | |
|---|---|---|
| result_id | uuid | PK, FK |
| datum_key | text | PK |
| label | text | |
| value | double precision | |
| citation_count | integer | |
| citation_source | text (trials\|api_query) | |
| source_query | jsonb | |
| citations_materialized | boolean | |

1 data point : N citations

**result_citations (keyset-paginated)**

| | | |
|---|---|---|
| result_id | uuid | PK, FK |
| datum_key | text | PK, FK |
| rank | integer | PK |
| nct_id | char(11) | |
| title | text | |
| evidence | jsonb [{field, excerpt}] | |

**jobs (background tasks)**

| | | |
|---|---|---|
| job_id | uuid | PK |
| request_key | char(64) | UQ (active) |
| request | jsonb | |
| plan | jsonb | |
| status | text (pending\|running\|succeeded\|failed\|cancelled) | idx |
| progress | jsonb | |
| estimated_seconds | integer | |
| result_id | uuid (nullable) | FK |
| error | jsonb | |
| attempts / max_attempts | integer | |
| locked_by | text | |
| heartbeat_at | timestamptz | |
| trace_id | uuid | |
| created_at / started_at / finished_at | timestamptz | |
| expires_at | timestamptz | |

**traces (observability)**

| | | |
|---|---|---|
| trace_id | uuid | PK |
| kind | text (request\|job) | |
| created_at | timestamptz | idx |
| request | jsonb | |
| http_status | integer | |
| outcome | text | |
| guard | jsonb | |
| stages | jsonb | |
| api_calls | jsonb | |
| result_id | uuid (nullable) | FK |
| job_id | uuid (nullable) | FK |

**llm_usage (one row per LLM call)**

| | | |
|---|---|---|
| call_id | uuid | PK |
| trace_id | uuid (nullable) | FK |
| job_id | uuid (nullable) | FK |
| stage | text (guard\|planner\|narrator) | |
| model | text | idx |
| attempt | integer | |
| is_fallback | boolean | |
| status | text (ok\|error\|timeout) | |
| input_tokens | integer | |
| cached_input_tokens | integer | |
| output_tokens | integer | |
| reasoning_tokens | integer | |
| latency_ms | integer | |
| cost_usd | numeric(12,6) | |
| price_version | text | |
| created_at | timestamptz | idx |